

The missing L^∞ endpoint for complex interpolation of the atomic variation spaces R^s

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Lane 13, model GPT5.6

11 August 2026

Status. Candidate complete endpoint theorem, likely valid, subject to human review. This removes the condition $q_1 \neq \infty$ from the two R^s inclusions in Theorem 4.4 of Amenta–Lorist–Veraar. It is classified as a substantial partial result relative to the paper’s broader interpolation questions: it does not settle the zero-loss V^s inclusion or the reverse R^s inclusion.

1 Source statement and endpoint left out

For a complex Banach space Y , a bounded interval J , and $1 \leq s < \infty$, Amenta–Lorist–Veraar define an $R^s(J; Y)$ -atom to be a function

$$a = \sum_{I \in \mathcal{I}} c_I \mathbf{1}_I, \quad \left(\sum_{I \in \mathcal{I}} \|c_I\|_Y^s \right)^{1/s} \leq 1, \quad (1)$$

where $\mathcal{I}$ is a family of mutually disjoint subintervals of J . The space $R^s(J; Y)$ consists of the L^∞ -convergent sums $f = \sum_k \lambda_k a_k$, with norm the infimum of $\sum_k |\lambda_k|$ over all such representations. Their endpoint convention is

$$R^\infty(J; Y) := L^\infty(J; Y). \quad (2)$$

Writing $q_\theta = [q_0, q_1]_\theta$, Theorem 4.4 of the source proves

$$R^{q_\theta}(J; Y) \hookrightarrow [R^{q_0}(J; Y), R^{q_1}(J; Y)]_\theta \quad (3)$$

and its ordered multi-interval R_0^s analogue, but in both statements adds the restriction $q_1 \neq \infty$ [1]. The page reproduced below also records the paper’s separate open questions about the V^s -scale and reverse inclusions.

Lemme 2] for scalar functions, and the argument extends to the general case. Part (ii) is straightforward to check. $\square$

We end this section with complex interpolation containments for the V^s - and R^s -classes. It is an open problem whether complex interpolation of the V^s -classes as below can be proved with $\varepsilon = 0$ (see [43, Chapter 12]). It is also not clear whether converse inclusions hold, but since we don’t need them we leave the question open.

Theorem 4.4. *Suppose $1 \leq q_0 \leq q_1 \leq \infty$, $\theta \in (0, 1)$, $\varepsilon > 0$ and let Y be a Banach space. Then for all bounded intervals $J \subset \mathbb{R}$ we have continuous inclusions*

$$(4.1) \quad V^{[q_0, q_1]_\theta - \varepsilon}(J; Y) \hookrightarrow [V^{q_0}(J; Y), V^{q_1}(J; Y)]_\theta,$$

$$(4.2) \quad R^{[q_0, q_1]_\theta}(J; Y) \hookrightarrow [R^{q_0}(J; Y), R^{q_1}(J; Y)]_\theta, \quad q_1 \neq \infty.$$

Furthermore, if $\mathcal{J} = (J_k)_{k \in \mathbb{N}}$ is an ordered collection of mutually disjoint bounded intervals in $\mathbb{R}$, then we have continuous inclusions

$$(4.3) \quad V_0^{[q_0, q_1]_\theta - \varepsilon}(\mathcal{J}; Y) \hookrightarrow [V_0^{q_0}(\mathcal{J}; Y), V_0^{q_1}(\mathcal{J}; Y)]_\theta$$

$$(4.4) \quad R_0^{[q_0, q_1]_\theta}(\mathcal{J}; Y) \hookrightarrow [R_0^{q_0}(\mathcal{J}; Y), R_0^{q_1}(\mathcal{J}; Y)]_\theta, \quad q_1 \neq \infty.$$

Proof. For $q_0 = 1$ and $q_1 = \infty$ we have (4.1) by applying subsequently [43, Lemma

Source context. Theorem 4.4 on page 13, including the two $q_1 \neq \infty$ restrictions and the adjacent interpolation questions.

2 Result

Theorem 1 (single-interval endpoint). *Let Y be a complex Banach space, let $J \subset \mathbb{R}$ be a bounded interval, let $1 \leq q < \infty$, and let $0 < \theta < 1$. Put*

$$\frac{1}{r} = \frac{1-\theta}{q}, \quad \text{equivalently} \quad r = \frac{q}{1-\theta}.$$

Then the missing endpoint inclusion holds:

$$R^r(J; Y) \hookrightarrow [R^q(J; Y), L^\infty(J; Y)]_\theta. \quad (4)$$

With the standard norms for the complex method and vector-valued sequence spaces, the inclusion in (4) is contractive.

Theorem 2 (ordered multi-interval endpoint). *Let $\mathcal{J} = (J_k)_{k \in \mathbb{N}}$ be an ordered family of mutually disjoint bounded intervals. Under the hypotheses and notation of Theorem 1,*

$$R_0^r(\mathcal{J}; Y) \hookrightarrow [R_0^q(\mathcal{J}; Y), R_0^\infty(\mathcal{J}; Y)]_\theta. \quad (5)$$

Here R_0^∞ is the c_0 -sum of the intervalwise L^∞ spaces, exactly as prescribed by the source definition.

Corollary 3. *The two R^s assertions (4.2) and (4.4) of Theorem 4.4 in [1] remain true for every $1 \leq q_0 \leq q_1 \leq \infty$; the displayed condition $q_1 \neq \infty$ can be deleted.*

3 Proof intuition

An atom in R^r already comes with precisely the data needed for interpolation: its coefficient vector lies in $\ell^r(Y)$, while the same step-function synthesis map sends $\ell^q(Y)$ into R^q and $\ell^\infty(Y)$ into L^∞ . Interpolating this one linear map places every R^r -atom in the desired complex interpolation space with uniform norm. The outer atomic norm is an ℓ^1 -sum, so summing the atoms completes the proof. The multi-interval statement then follows by the standard fact that complex interpolation commutes with c_0 -sums.

The only point requiring care is identification of the limit: the atomic series is given to converge in L^∞ , while the proof also makes it converge in the interpolation space. Both limits agree because the interpolation space embeds continuously into L^∞ .

4 Proofs

Proof of Theorem 1. Fix a family $\mathcal{I}$ of mutually disjoint subintervals of J , and define its synthesis operator

$$T_{\mathcal{I}}c := \sum_{I \in \mathcal{I}} c_I \mathbf{1}_I. \quad (6)$$

For $c \in \ell^q(\mathcal{I}; Y)$, the normalized function $T_{\mathcal{I}}c/\|c\|_{\ell^q}$ is an R^q -atom (unless $c = 0$). Consequently

$$\|T_{\mathcal{I}} : \ell^q(\mathcal{I}; Y) \rightarrow R^q(J; Y)\| \leq 1. \quad (7)$$

Disjointness also gives the endpoint estimate

$$\|T_{\mathcal{I}}c\|_{L^\infty(J; Y)} \leq \|c\|_{\ell^\infty(\mathcal{I}; Y)}, \quad (8)$$

so $T_{\mathcal{I}} : \ell^\infty(\mathcal{I}; Y) \rightarrow L^\infty(J; Y)$ has norm at most one.

The complex interpolation theorem for operators, together with the standard vector-valued sequence-space identity

$$[\ell^q(\mathcal{I}; Y), \ell^\infty(\mathcal{I}; Y)]_\theta = \ell^r(\mathcal{I}; Y), \quad \frac{1}{r} = \frac{1-\theta}{q}, \quad (9)$$

therefore yields

$$\|T_{\mathcal{I}}c\|_{[R^q(J; Y), L^\infty(J; Y)]_\theta} \leq \|c\|_{\ell^r(\mathcal{I}; Y)}. \quad (10)$$

Identity (9) is valid for arbitrary counting index sets; alternatively, each $c \in \ell^r$ has countable support and is the ℓ^r -limit of finite-support vectors, so the usual finite-dimensional identity and completion suffice. In particular, every R^r -atom has interpolation norm at most one.

Now take $f \in R^r(J; Y)$ and an atomic representation

$$f = \sum_{k=1}^{\infty} \lambda_k a_k, \quad (\lambda_k) \in \ell^1, \quad (11)$$

converging in $L^\infty(J; Y)$. By (10), the partial sums in (11) are Cauchy in $X_\theta := [R^q(J; Y), L^\infty(J; Y)]_\theta$ and converge there to some g , with

$$\|g\|_{X_\theta} \leq \sum_{k=1}^{\infty} |\lambda_k|. \quad (12)$$

The identity maps both $R^q(J; Y)$ and $L^\infty(J; Y)$ contractively into $L^\infty(J; Y)$, so operator interpolation gives a contractive embedding $X_\theta \hookrightarrow L^\infty(J; Y)$. Thus the same partial sums converge in L^∞ to g . Their prescribed L^∞ -limit is f , hence $g = f$. Taking the infimum in (12) over all atomic representations proves (4). $\square$

Proof of Theorem 2. Affine reparametrization of each J_k onto a fixed bounded interval J_* is an isometry for every R^u -norm, including $u = \infty$. Therefore

$$R_0^u(\mathcal{J}; Y) \cong c_0(\mathbb{N}; R^u(J_*; Y)) \quad (1 \leq u \leq \infty) \quad (13)$$

isometrically. Complex interpolation commutes with c_0 -sums [3, Remark 3, Section 1.18.1]. Applying Theorem 1 coordinatewise, with its constant independent of the interval, gives

$$\begin{aligned} R_0^r(\mathcal{J}; Y) &\cong c_0(\mathbb{N}; R^r(J_*; Y)) \\ &\hookrightarrow c_0(\mathbb{N}; [R^q(J_*; Y), L^\infty(J_*; Y)]_\theta) \\ &= [c_0(\mathbb{N}; R^q(J_*; Y)), c_0(\mathbb{N}; L^\infty(J_*; Y))]_\theta \\ &\cong [R_0^q(\mathcal{J}; Y), R_0^\infty(\mathcal{J}; Y)]_\theta. \end{aligned}$$

This is (5). $\square$

5 Verification and scope

The proof uses only three exact ingredients: the two endpoint bounds (7)–(8), interpolation of vector-valued ℓ^p spaces, and interpolation of c_0 -sums. Infinite atomic families cause no measurability or convergence problem: an ℓ^r coefficient vector has countable support, and the synthesis operator is bounded on the whole compatible endpoint couple. The outer atomic series is controlled in the interpolation norm by the same ℓ^1 coefficient sum used in the definition of R^r .

This packet does *not* prove $V^{q_\theta} \hookrightarrow [V^{q_0}, V^{q_1}]_\theta$ without an epsilon loss. It also does not prove $[R^{q_0}, R^{q_1}]_\theta \hookrightarrow R^{q_\theta}$. The reverse V^s inclusion, which was described as unclear in the 2017 source, is explicitly proved by an increment-operator argument in a 2026 follow-up [4, Lemma 2.4]; it is therefore not claimed here.

6 Novelty audit and reviewer focus

The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:1705.07792, the exact paper title, R^s , bounded variation, the $q_1 = \infty$ endpoint, and complex interpolation. Exact and close full-text searches of the local arXiv source corpus found only the source theorem. Bounded web searches on 11 August 2026 used the same terms and inspected the 2026 follow-up by Deng–Lorist–Veraar. That paper proves the reverse V^s inclusion and continues to cite the 2017 paper for the opposite-direction R^s result; no statement of the endpoint theorem (4)–(5) was located.

Novelty confidence is moderate rather than high because the proof is a short direct consequence of the atomic definition and standard complex interpolation. The most useful human check is whether the original restriction concealed a convention not visible in Definition 4.2. Under the printed definitions, the synthesis argument applies verbatim. A reviewer should also check the standard c_0 -sum identity at the L^∞ endpoint; this is exactly the identity already invoked in the source's proof of its multi-interval statements.

References

- [1] A. Amenta, E. Lorist, and M. Veraar, *Fourier multipliers in Banach function spaces with UMD concavifications*, Trans. Amer. Math. Soc. 371 (2019), 4837–4868; arXiv:1705.07792.
- [2] J. Bergh and J. Löfström, *Interpolation Spaces: An Introduction*, Springer, 1976.
- [3] H. Triebel, *Interpolation Theory, Function Spaces, Differential Operators*, North-Holland, 1978.
- [4] C. Deng, E. Lorist, and M. Veraar, *Operator-valued Fourier multipliers of bounded s -variation*, arXiv:2601.04803, 2026.